Fiscal Year:	FY 2011	**Task Last Updated:**	FY 06/28/2011
PI Name:	Bassham, Diane Ph.D.		
Project Title:	Molecular Basis for the Role of the Vacuole in Shoot Gravitropism		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	50011-0001	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Space Biology--Microbial, Plant, Cell Biology: NNH08ZTT003N
Start Date:	08/01/2009	**End Date:**	07/31/2013
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA KSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: end date changed to 7/31/2013, per NSSC information (Ed. 5/8/2012)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX09AK78G		
Performance Goal No.:			
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Task Description:	Plant perception and response to gravity is critical for appropriate growth, development and environmental responses. Shoot gravitropism involves the perception of gravity by sedimentation of starch-containing amyloplasts in gravity-sensing cells, followed by transmission of the signal to additional cell types to promote directional growth. Vesicle trafficking to the vacuole is essential for the perception of the gravity signal, but the molecular basis for this requirement is unknown. Our long term goal is to determine the mechanism by which gravity is perceived by shoots and the gravity signal transduced to other cell types to produce a response. The objective of this application is to determine the function of vacuolar trafficking proteins in shoot gravitropism. We hypothesize that appropriate physical properties of the vacuole are essential for amyloplast sedimentation, and therefore for gravity perception. We will test this hypothesis by measuring the elasticity of wild type and mutant vacuoles using atomic force microscopy and by comparing the lipid constituents of the vacuolar membranes. We will also identify additional proteins involved in vacuolar trafficking by a combination of in vitro membrane fusion assays and in vivo immunoprecipitations and test their function in gravitropism using genetic approaches. This research will provide a molecular explanation for the observation that shoot gravitropism is specifically defective in vacuolar trafficking mutants and allow us to make testable predictions about the mechanistic basis for their agravitropic phenotype. It will also provide a foundation for the generation of transgenic plants with altered gravity response thresholds.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The primary impact of our research will be to increase understanding of the mechanisms by which plants respond to gravity. In addition to being important for the growth of plants in the altered gravitational conditions likely to be encountered during space flight, this research will also contribute to our knowledge of plant growth, development and response to environmental cues. The gravitational response interacts extensively with other developmental, hormonal and environmental response pathways, leading to morphological and anatomical changes to the plant structure. Understanding of the molecular mechanism of gravitropism is therefore vital for the development of strategies to improve plant growth and yield under varying conditions.
Task Progress:	Personnel A PhD rotation student, Rahul Roy, has been recruited to work on this project full time and the research will form the basis of his PhD thesis. Another graduate student, Sang-Jin Kim, participated in the project and has now graduated with his PhD. Sang-Jin established assays to measure vesicle fusion and identified and characterized putative vacuolar trafficking proteins. Research progress Identification of mutants in potential shoot gravitropism genes. Our working hypothesis is that vacuolar trafficking proteins VTI11 and SYP22, both of which are required for gravitropism, are part of a SNARE complex that delivers membrane and cargo to the vacuole. To identify new vacuolar trafficking proteins that may function in gravity responses, co-immunoprecipitation was performed to isolate proteins that interact with the known vacuolar trafficking protein SYP41. A previously uncharacterized protein, named TNO1, was identified as a SYP41-interacting factor. Vacuolar proteins were partially secreted to the apoplast in a tno1 mutant, suggesting that TNO1 is required for efficient protein trafficking to the vacuole. The function of TNO1 in gravitropism in both hypocotyls and inflorescence stems is now underway. Based in previous immunoprecipitation results, we predict that the additional trafficking proteins SYP51 and/or SYP52 may be present in the same complex and therefore may also be required for shoot gravitropism. To test this hypothesis, Arabidopsis mutants with T-DNA insertions in the SYP51 and SYP52 genes have been identified by genomic PCR using primers specific to the SYP51 and SYP52 genes and to the TDNA borders. RT-PCR analysis indicates that homozygous syp52 mutants have lost expression of the SYP52 gene and gravitropism assays are now being performed to determine the effect on gravity responses. No homozygous mutants for SYP51 have been identified, suggesting that this is an essential gene. Analysis of membrane fusion activity of the potential vesicle fusion proteins To biochemically analyze the function of putative vesicle fusion proteins involved in vacuolar trafficking, in vitro liposome fusion assays are being performed using recombinant proteins to determine whether these proteins are able to catalyze membrane fusion as predicted. Constructs have been generated in E. coli expression vectors for each protein, fused to a 6xHis-tag to allow protein purification. Several of these proteins have been successfully expressed in large quantities in E. coli and affinity purified to remove unwanted contaminants. These purified proteins are being reconstituted into artificial proteoliposomes and assayed for their ability to fuse vesicles. Several of the proteins that are thought to be involved in vacuolar trafficking based on their mutant phenotypes have been analyzed in this way and shown to be sufficient to catalyze the fusion of liposomes. Additional factors are now being analyzed in the same way.
Bibliography Type:	Description: (Last Updated: 09/18/2019)
Articles in Peer-reviewed Journals	Kim SJ, Bassham DC. "TNO1 is involved in salt tolerance and vacuolar trafficking in Arabidopsis." Plant Physiol. 2011 Jun;156(2):514-26. Epub 2011 Apr 26. PMID:21521696 , Jun-2011
Articles in Peer-reviewed Journals	Peng J, Ilarslan H, Wurtele ES, Bassham DC. "AtRabD2b and AtRabD2c have overlapping functions in pollen development and pollen tube growth." BMC Plant Biol. 2011 Jan 26;11:25. PMID:21269510 , Jan-2011
Dissertations and Theses	Kim SJ. "Role of TNO1 and vesicle fusion machineries at the Arabidopsis trans-Golgi network." Dissertation, Iowa State University, August 2010. , Aug-2010